

NADEN MOHAMED YASEN

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SUMMARY

AI & Computer Science graduate (Helwan University, 2026) with hands-on experience building, containerizing, and monitoring ML/GenAI systems in production-like environments. Comfortable across the MLOps lifecycle — from designing database schemas and managing migrations, to containerizing multi-service systems with Docker, to instrumenting APIs with Prometheus/Grafana for real-time observability. Strong foundation in Python, SQL, and ML/statistics, with hands-on deep learning and GenAI experience (PyTorch, TensorFlow, LLMs, fine-tuning). Comfortable diagnosing production-grade issues independently through systematic, log-driven debugging.

EDUCATION

B.Sc. Computer Science & Artificial Intelligence · Helwan University, Cairo · GPA: 3.37/4.0 *Sep 2022 – Jun 2026*

SKILLS

Programming & Databases: Python, SQL, C, C++, PostgreSQL, pgvector, SQLAlchemy, Alembic, MongoDB Atlas, Qdrant, FAISS

MLOps / DevOps: Docker, Docker Compose (multi-service orchestration), Prometheus, Grafana, Nginx, Git/GitHub, database migrations, containerized deployment workflows, API monitoring & observability

ML & Statistics: Model Evaluation & Benchmarking, Statistics, Linear Algebra, Scikit-learn, XGBoost, Exploratory Data Analysis

Deep Learning & GenAI: PyTorch, TensorFlow, LLMs, RAG, Fine-Tuning (LoRA/PEFT), NLP, Agentic Systems, Hugging Face, LangChain, LangGraph

Frameworks & Tools: FastAPI, Jupyter, Postman, pytest

Languages Spoken: Arabic, English

RELEVANT PROJECTS

ScholarFlow AI - RAG-based Document Intelligence Platform | *Python, FastAPI, PostgreSQL/pgvector, Qdrant, Docker*

- Designed and built a RAG backend with a provider-factory architecture, allowing the LLM (Groq/Gemini) and vector database (pgvector/Qdrant) to be swapped via configuration with no code changes.
- Designed the PostgreSQL schema and managed migrations (Alembic) for projects, uploaded assets, and vector-indexed chunks.
- Containerized the full system with Docker Compose across 7 services (FastAPI, PostgreSQL/pgvector, Qdrant, Nginx, Prometheus, Grafana, exporters), including a custom migration-then-serve endpoint workflow.
- Instrumented the API with Prometheus metrics (latency, throughput) and Grafana dashboards, then extended the stack with RAG-specific metrics (empty-retrieval rate, retrieval confidence distribution) to monitor answer quality in real time, not just uptime. Designed the PostgreSQL schema, managed migrations, and containerized the full 7-service system with Docker Compose; independently debugged and fixed production-grade issues across the stack.
- Built an offline evaluation harness combining deterministic retrieval metrics (hit rate, MRR) with RAGAS-based generation metrics (faithfulness, answer relevancy) to catch quality regressions before deployment.
- Independently diagnosed and resolved production-grade issues — container networking misconfigurations, a silent vector-insertion bug, and an API logic error — using a systematic workflow of container inspection, log tracing, and root-cause isolation.

Adaptive Code Intelligence Assistant | *LLM Evaluation, Pydantic, pytest*

- Designed a multi-step LLM pipeline with strict Pydantic schemas at every stage, keeping component handoffs reliable and testable.
- Developed an evaluation harness to measure router classification accuracy and end-to-end code-generation correctness (pass@1) against a benchmark dataset, transforming LLM output quality into trackable metrics.

Toxic Content Classification System | *PyTorch, DistilBERT, LoRA, Llama Guard, BLIP*

- Evaluated and compared modeling approaches (LSTM baseline vs. LoRA-fine-tuned DistilBERT), tracking accuracy, precision, recall, and F1 across categories using Weights & Biases.
- Tested a zero-shot content moderation pipeline (Llama Guard) alongside a multimodal BLIP captioning module, weighing trade-offs between approaches ahead of Streamlit deployment.

TRAINING & CERTIFICATIONS

IBM-Certified AI & Data Science Professional | Digital Egypt Pioneers Initiative (DEPI) *Nov 2024 – May 2025*

- Completed an IBM-certified program covering Python, ML, SQL, and MLOps; built and evaluated ML/DL models, achieving 91.12% accuracy on the capstone project.

NLP Training | Neurova AI *Apr 2026 – Jul 2026*

- Completed an intensive 8-week NLP training program spanning classical preprocessing, TF-IDF, Transformer architectures, BERT fine-tuning, RAG, AI Agents and LLM integration across 7 end-to-end projects.